KYLE T. DAVID

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CURRENT APPOINTMENTS

Postdoctoral Fellow | Rokas Lab, Vanderbilt University — 2022 - Present
Department of Biological Sciences

Postdoctoral Fellow | Pennell Lab, Cornell University — 2026 - Present
Department of Computational Biology

EDUCATION

PhD Biology | Halanych Lab, Auburn University — 2017 - 2022
Department of Biological Sciences

BS Biology & Marine Science | University of Miami — 2012 - 2016
Rosenstiel School of Marine & Atmospheric Science, Chemistry Minor

PUBLICATIONS

19. **David, K. T.**, Horianopoulos, L., Gonçalves, C., Steenwyk, J. L., Pontes, A., Gonçalves, P., Hittinger, C. T., Pennell, M., & Rokas, A. (2026). **Discovery of additional ancient genome duplications in yeasts.** *Current Biology*, 36(4), 1067-1075.

18. Riedling, O. L., **David, K. T.**, & Rokas, A. (2025). **Global patterns of diversity and distribution in Aspergillus fungi are driven by human and environmental influences**. *Current Biology,* 35(18), 4453-4466.

17. **David, K. T**., Schraiber, J. G., Crandall, J. G., Labella, A. L., Opulente, D. A., Harrison, M. C., ... & Rokas, A. (2025). **Convergent expansions of keystone gene families drive metabolic innovation in Saccharomycotina yeasts**. *Proceedings of the National Academy of Sciences*, 122(23), e2500165122.

16. Feng, B., Li, Y., Xu, B., Liu, H., Steenwyk, J. L., **David, K. T.**, ... & Li, Y. (2025). **Unique trajectory of gene family evolution from genomic analysis of nearly all known species in an ancient yeast lineage**. *Molecular Systems Biology*, 21(8), 1066-1089.

15. Warner, J.F., Range, R.C., Fenner, J., Ka, C., Waits, D.S., Boddy, K., **David, K.T.**, Mahon, A.R. and Halanych, K.M. (2024). **Chromosomal-Level Genome Assembly of the Antarctic Sea Urchin *Sterechinus neumayeri*: A Model for Antarctic Invertebrate Biology**. *Genome Biology and Evolution*, *16*(11), evae237.

14. **David, K. T.**, Harrison, M. C., Opulente, D. A., LaBella, A. L., Wolters, J. F., Zhou, X., ... & Rokas, A. (2024). **Saccharomycotina yeasts defy longstanding macroecological patterns.** *Proceedings of the National Academy of Sciences, 121*(10), e2316031121.

13. Sun, L., **David, K. T.**, Wolters, J. F., Karlen, S. D., Gonçalves, C., Opulente, D. A., ... & Hittinger, C. T. (2024) Functional and evolutionary integration of a fungal gene with a bacterial operon. *Molecular Biology and Evolution*, msae045.

12. **David, K. T.**, & Halanych, K. M. (2023). Unsupervised deep learning can identify protein functional groups from unaligned sequences. *Genome Biology and Evolution*, evad084.

11. **David, K. T.** (2022). Global gradients in the distribution of animal polyploids. *Proceedings of the National Academy of Sciences*, *119*(48), e2214070119.

10. Tassia, M. G., **David, K. T.**, Townsend, J. P., & Halanych, K. M. (2021). TIAMMAt: Leveraging biodiversity to revise protein domain models, evidence from innate immunity. *Molecular Biology and Evolution*, *38*(12), 5806-5818.

9. **David, K. T.**, & Halanych, K. M. (2021). Spatial proximity between polyploids across South American frog genera. *Journal of Biogeography*, *48*(5), 991-1000.

8. Zhang, Y., Yap, K. N., **David, K. T.**, & Swanson, D. L. (2021). The high-energy aerial insectivore lifestyle of swallows does not produce clear thermogenic side effects. *Ornithology*, *138*(3), ukab022.

7. Li, Y., **David, K. T.**, Shen, X. X., Steenwyk, J. L., Halanych, K. M., & Rokas, A. (2020). Feature frequency profile-based phylogenies are inaccurate. *Proceedings of the National Academy of Sciences*, *117*(50), 31580-31581.

6. **David, K. T.**, Oaks, J. R., & Halanych, K. M. (2020). Patterns of gene evolution following duplications and speciations in vertebrates. *PeerJ*, *8*, e8813.

5. Li, Y., Tassia, M.G., Waits, D.S., Bogantes, V. E., **David, K. T.**, & Halanych, K.M. (2019). Genomic adaptations to chemosymbiosis in the deep-sea seep-dwelling tubeworm *Lamellibrachia luymesi*. *BMC Biology,* 17, 91.

4. **David, K. T.**, Wilson, A. E., & Halanych, K. M. (2019). Sequencing disparity in the genomic era. *Molecular Biology and Evolution, 36*(8), 1624-1627.

3. **David, K. T.**, & Halanych, K. M. (2017). Mitochondrial genome of *Dinophilus gyrociliatus* (Annelida: Dinophilidae). *Mitochondrial DNA Part B*, 2(2), 831-832.

2. McMahan, C. D., Ginger, L., Cage, M., **David, K. T.**, Chakrabarty, P., Johnston, M., & Matamoros, W. A. (2017). Pleistocene to Holocene expansion of the black-belt cichlid in Central America, *Vieja maculicauda* (Teleostei: Cichlidae). *PLOS ONE*, *12*(5), e0178439.

1. **David, K. T.**, Tanabe, P., & Fieber, L. A. (2016). Resource availability drives mating role selection in a simultaneous hermaphrodite *Aplysia californica*. *The Biological Bulletin*, *231*(3), 199-206.

FELLOWSHIPS

Anthropic Claude for Science Cohort - $10,000 — 2026

NSF Postdoctoral Research Fellowship in Biology - $240,000 — 2023-2026

NSF Graduate Research Fellowship - $138,000 — 2019-2022

Auburn Cellular & Molecular Biosciences Fellowship - $22,500 2017-2018

AWARDS & HONORS

Vanderbilt Postdoctoral Fellow of the Year Award - $500 2026

Evolutionary Studies Excellence in Outreach Award 2025

American Society of Naturalists Early Career Investigator Award - $1,400 2024

W. D. Hamilton Award, Finalist - $500 2022

Antarctica Service Medal 2021

Editors' Choice | *Journal of Biogeography* 2021

PacBio Plant and Animal Sciences SMRT Grant, Semifinalist 2020

Cover Article | *Molecular Biology and Evolution* 2019

Wake Award for Best Student Oral Presentation - $150 2019

INVITED TALKS

12. Genome rearrangements set the breadth and rate of metabolic variation across yeasts. Vanderbilt Postdoctoral Association Seminar. November 2026.
11. Metabolic consequences of genome rearrangements across yeasts. University of Maryland. October 2026.
10. Genome duplication and the onset of ecological novelty. *Evolution* meeting in Cleveland, Ohio. June 2026.
9. Genome architecture and the evolution of novelty. Vanderbilt University. May 2026.
8. The role of genetic novelty in environmental adaptation. Cornell University. March 2026.
7. Using big data to answer big questions in evolution and ecology. University of North Carolina Charlotte. September 2025.
6. Evolution of novelty. *Evolution* meeting in Montreal, Canada. July 2024.
5. Macroecological consequences of whole-genome duplication: a latitudinal polyploid gradient in three animal clades. *Society for Experimental Biology m*eeting in Prague, Czech Republic. July 2024.
4. Responsible research conduct in bioinformatics: a case study in the cancer microbiome. Vanderbilt University. January 2024.

3. **Global gradients in the distribution of animal polyploids**. *Plant & Animal Genomes* conference in San Diego, California. January 2023.
2. **Environment and ploidy in South American frogs**. Polyploid Webinar organized by the University of Arizona. September 2021.
1. **Evolutionary consequences of gen(om)e duplication in animals.** Vanderbilt University. August 2021.

PRESENTATIONS

10. **Discovery of two ancient genome duplications that shaped Saccharomycotina yeast evolution.** Talk presented at the *Evolution* meeting in Athens, Georgia. June 2025.
9. **Convergent genome evolution underlies metabolic and ecological innovation in Saccharomycotina.** Poster presented at the *Fungal Genetics* meeting in Monterey Bay, California. March 2024.
8. **The yeast subphylum Saccharomycotina defies longstanding macroecological patterns.** Talk presented at the *Evolution* meeting in Albuquerque, New Mexico. June 2023.
7. **Global gradients in the distribution of animal polyploids.** Talk presented at the *Evolution* meeting in Cleveland, Ohio. June 2022.
6. **Bad role models: can deep learning save us from the tyranny of mouse, worm, and fly?** Talk presented at the virtual *Evolution* meeting. June 2021.
5. **Are two genomes better than one? Ploidy correlated species' distributions in South American frogs.** Talk presented at the *Society for Integrative and Comparative Biology* meeting in Austin, Texas. January 2020.
4. **Sequencing disparity in the genomic era.** Talk presented at the *Evolution* meeting in Providence, Rhode Island. June 2019.
3. **Much ado about orthologs: consequences of duplication and speciation in gene evolution.** Talk presented at the *Society for Integrative and Comparative Biology* meeting in Tampa, Florida. January 2019.
2. **Orthologs vs. paralogs, what's the difference?** Talk presented at the *Society of Systematic Biologists* standalone meeting in Columbus, Ohio. June 2018.
1. **Mating role choice in a simultaneous hermaphrodite.** Poster presented at the *Undergraduate Research, Creativity, and Innovation Forum* in Miami, Florida. June 2016.

TEACHING

Certificate in College Teaching | Vanderbilt University 2024

I have a teaching certificate from Vanderbilt's Center for Teaching, having completed the required 16-week combined seminar and practicum.

Guest Lecturer | University of West Florida 2023, 2026
PCB4673: Principles of Evolution

Guest Lecturer | Vanderbilt University 2022
BSCI 3890/6890: History of Life

Lecturer | Julia Tutwiler Prison for Women 2022
Prepared and presented a lecture and lesson plan for Sparks Science and Mathematics Series, collaborative effort to bring STEM programming to people who are currently incarcerated in Alabama.

Guest Lecturer | Brooklyn College 2021
MHC 20301: "Science Forward" seminar course

Co-Instructor | Auburn University 2020
BIOL 3030: Evolution and Systematics

Guest Lecturer | Staton Correctional Facility 2019
Prepared and presented a lecture as part of Auburn University's Alabama Prison Arts + Education Project.

Teaching Assistant | Auburn University 2019
BIOL 1031: Organismal Biology

Teaching Assistant | Auburn University 2018
BIOL 4010: Invertebrate Biodiversity

SERVICE & OUTREACH

Interviews
Quanta Magazine, All Things Considered, Hakai Magazine, Tiny Living Beings podcast

Reviews
Genome Biology and Evolution, *Evolution Letters*, *Nature Ecology & Evolution*, *Ecology Letters*, *Molecular Ecology*, *Journal of Heredity*, *Systematic Biology*, *Journal of Molecular Evolution*, *Journal of Biogeography*, *Oikos*, *PeerJ*, *Mitochondrial DNA Part B: Resources*

Invited Panels
Vanderbilt Undergraduate Research Symposium 2026
NSF Systematics and Biodiversity Science Panel 2025

W. D. Hamilton Award, *Evolution* 2023
Undergraduate Travel Award, *The Allied Genetics Conference* 2023 & 2024
Envision Ecology, Evolution, and Behavior Graduate Preview Weekend at Michigan State University 2021

Evolutionary Studies Initiative, Outreach Co-coordinator 2022 - Present
An interdepartmental initiative at Vanderbilt engaged in evolutionary research, education, and outreach. Participated in multiple outreach events, as well as developed and led interactive teaching modules at K-12 schools in underserved communities.

Labs Behind Bars, Committee Member 2023 - 2025
Served on the committee for Labs Behind Bars, a Clinton Global Initiative Project to adapt and develop STEM labs from Vanderbilt's curricula to a prison setting.

Dismas House 2023 - 2025

Developed, organized, and operated a collaboration between the Evolutionary Studies Initiative at Vanderbilt and the Dismas House residential reentry program for Tennessee state correctional facilities. Dismas House residents are regularly hosted on Vanderbilt's campus to tour facilities and participate in educational programming and outreach.

Skype a Scientist 2018 - 2022

Engaged in ~60 minute Q&A and lecture sessions with K-12 classrooms virtually.

Summer Science Institute 2021

Counselor for Auburn University's Summer Science Institute, a weeklong sleepaway camp for 11th and 12th grade students with an aptitude and interest in science and math.

Auburn University Museum of Natural History 2017 - 2022

Participated in several engagement opportunities through the museum including open houses and outreach events with underprivileged schools.

Curious Curators Camp 2019

Counselor for 5th-6th grade summer camp that teaches the importance of natural history museums. Oversaw educational exercises and activities.

Field Museum of Natural History 2016 - 2017

Participated in "Talk to a Scientist Hour" and "Meet a Scientist" programs while interning at the Field Museum's DNA Discovery Center.

SELECTED FIELD EXPERIENCE

Costa Rica 2022

7 days at La Selva Biological Station and 5 days at Palo Verde Research Station as part of advanced coursework.

Antarctic Peninsula & Weddell Sea 2020

84 days on the *RVIB Nathaniel B. Palmer* sampling benthic invertebrate diversity.

Galápagos Islands 2015

~75 days on Isabela and surrounding islands for study abroad.

REFERENCES

Dr. Antonis Rokas	**Dr. Matt Pennell**	**Dr. Ken Halanych**	**Dr. Jamie Oaks**
Professor	Associate Professor	Executive Director	Associate Professor
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Current Mentor	*Current Co-mentor*	*PhD Advisor*	*Interim PhD Advisor*

~Updated November 2026~